

Arithmetic ground-state geometry

Supersymmetric local factors and the positive-pairing problem

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Abstract

We investigate positive supersymmetric quantum theories as possible sources of an exact Hilbert-norm representation of the finite-interval Weil form. The complete positivity identity remains open. We first explain how a positive gamma kinetic energy and a finite-prime Morse model isolate the missing contact and pole contributions. We then construct a four-supercharge theory on a cylinder, retain the gamma function as a holomorphic period, and obtain a physical Euler-factor metric from quadratic prime fields. A cross-coupled quartic sector has nine physical vacua. At fixed quartic coupling, an exact differential equation excludes the Euler identity period on every admissible flat rapid-decay cycle. Scaling the quartic interaction with the square of the prime mass restores both the Euler period and its modulus-squared dependence in the physical identity-class metric. This interacting mass direction is nevertheless trivial in the Jacobi ring. Finite-rank curvature, scalar ground-transform, and normalization tests further restrict prospective completions. The results identify a concrete next problem: a physical pairing with nontrivial interaction geometry that fixes the source normalization, contact and poles together.

1 Introduction: the project and what has been learned

The purpose of this project is to explain an arithmetic positivity statement by constructing a system whose positivity is already understood. The arithmetic object is the Weil quadratic form associated with the Riemann zeta function. Its nonnegativity on every smooth compactly supported complex input is equivalent to the Riemann hypothesis. The desired argument would identify this form with an ordinary positive bulk energy or Hilbert-space norm. Numerical tests could then check the identity; they would not be the source of its sign.

This manuscript develops one route toward such a construction. It studies supersymmetric ground states, their parameter-dependent metrics, and interacting theories that retain prescribed gamma and prime factors. It produces exact local-factor identities and exclusions of specified repairs. It does not produce the full Weil norm identity. Understanding both the successes and their limits is essential: a positive quantum Hamiltonian can encode an arithmetic factor without making the required arithmetic quadratic form positive.

1.1 The arithmetic quantity that the theory must reproduce

Let f be a smooth complex function supported in $I_L = (-L/2, L/2)$, and let F be its zero extension to $\mathbb{R}$. Use the Fourier convention

$$\hat{F}(\tau) = \int_{\mathbb{R}} F(x) e^{-i\tau x} dx.$$

In the normalization used here, the target has the decomposition

$$\begin{aligned} Q_L[f] = & K[F] + w_0 \|f\|^2 + 2|C(f)|^2 - 2|S(f)|^2 \\ & - 2 \sum_{\substack{p \text{ prime}, m \geq 1 \\ m \log p < L}} (\log p) p^{-m/2} \operatorname{Re} \langle F, U_{m \log p} F \rangle. \end{aligned} \quad (1.1)$$

Here $U_d F(x) = F(x - d)$ compares the profile with a translated copy, and

$$C(f) = \int_{I_L} f(x) \cosh(x/2) dx, \quad S(f) = \int_{I_L} f(x) \sinh(x/2) dx.$$

The two squared global amplitudes constitute the signed pole contribution. The gamma kinetic energy and its local normalization are

$$\begin{aligned} K[F] &= \frac{1}{2\pi} \int_{\mathbb{R}} B(\tau^2) |\widehat{F}(\tau)|^2 d\tau, & w_0 &= \psi(1/4) - \log \pi < 0, \\ B(s) &= \operatorname{Re} \psi(1/4 + i\sqrt{s}/2) - \psi(1/4), & s &\geq 0, \end{aligned} \quad (1.2)$$

where $\psi = \Gamma'/\Gamma$ is the digamma function. The distances $m \log p$ and weights $(\log p)p^{-m/2}$ are fixed arithmetic data. A finite support length activates only finitely many such distances.

The gamma kinetic term is nonnegative, but the contact constant, translated correlations and pole contribution cannot be discarded. The project requires the complete sum in (1.1) for all inputs and arbitrarily large L , rather than a chosen finite collection of functions or a fixed list of primes at every length. The Weil criterion and modern operator formulations provide the mathematical setting [1]. Section 2 supplies the form domain and the identities used below; no separate background document is needed to read this manuscript.

1.2 What a positive bulk or quantum realization would mean

In a classical auxiliary-field construction, one fixes the source f , minimizes a nonnegative bulk energy over responding fields u , and seeks

$$Q_L[f] = \inf_u \mathcal{E}_L[f, u], \quad \mathcal{E}_L[f, u] \geq 0. \quad (1.3)$$

The bulk may consist of fields on an extra coordinate, a collection of channels, or other auxiliary degrees of freedom. Its defining action and source map must be specified independently of the unknown positivity of Q_L .

The operator version seeks a positive physical Hilbert space and a linear preparation map $\mathcal{A}_L$ with

$$Q_L[f] = \|\mathcal{A}_L f\|_{\mathcal{H}}^2. \quad (1.4)$$

Equivalently, the associated form operator would satisfy $W_L = \mathcal{A}_L^* \mathcal{A}_L$ in the appropriate form sense. This replaces separate tests of individual functions by one operator identity. The requirement that the identity hold on the full input domain remains. Defining $\mathcal{A}_L$ through an assumed positive square root of W_L would simply assume the desired result.

Supersymmetry offers a way to organize such a physical construction. A nilpotent charge Q with its actual Hilbert adjoint gives

$$H = QQ^\dagger + Q^\dagger Q \geq 0.$$

Its zero-energy states are harmonic representatives: states annihilated by both charges. Their ordinary norms can be nonzero even though their energies vanish. Thus the proposed observable in a supersymmetric ground-state construction is a prepared norm or boundary pairing, not merely the ground-state energy. A positive Hamiltonian alone does not establish (1.4). Quantizing an auxiliary system is useful only if the quantum state space, interactions or gluing law supplies an identity absent from the classical minimization.

1.3 The starting success and the remaining discrepancy

The gamma kinetic term already has an exact positive auxiliary-field realization. The identity

$$B(s) = \sum_{k \geq 0} \frac{2}{a_k} \frac{s}{s + a_k^2}, \quad a_k = 2k + \frac{1}{2},$$

expresses it as a positive tower of resolvent responses. Each response comes from minimizing a squared mismatch and a positive gradient energy. This is a substantial first success: an entire continuous operator, on its logarithmic form domain, is represented by a positive field energy. It does not include the contact or poles.

A second starting model is a positive supersymmetric Morse system tensored with discrete prime sectors. For a finite prime set $\mathcal{S}$ it has a state Ψ_σ and a self-adjoint observable X satisfying

$$\langle \Psi_\sigma, e^{i\tau X} \Psi_\sigma \rangle = \Lambda_{\mathcal{S}}(\sigma + i\tau), \quad \Lambda_{\mathcal{S}}(z) = \pi^{-z/2} \Gamma(z/2) \prod_{p \in \mathcal{S}} (1 - p^{-z})^{-1}. \quad (1.5)$$

The formula is verified by an elementary integral and geometric sums. Its normalized logarithmic derivative yields a positive gamma-plus-prime jump energy $J_{\mathcal{S},L}$. The complete target, however, is

$$W_L = J_{\mathcal{S},L} - \kappa_{\mathcal{S}} I + P_L, \quad \kappa_{\mathcal{S}} > 0, \quad (1.6)$$

when $\mathcal{S}$ includes all primes active on the interval. The missing negative contact and signed pole operator are therefore precise, rather than unspecified corrections. A positive characteristic function or jump process does not by itself account for them.

1.4 Why enlarge the theory and admit interactions?

The original arithmetic state family is too restrictive for the desired extended supersymmetry. Its chosen complex extension has a nonflat scalar metric, whereas an ordinary one-vacuum chiral tt^* system has zero curvature. Here tt^* denotes the compatibility equations relating a physical vacuum metric, its projected connection, and operators induced by variations of a holomorphic superpotential. More vacua permit matrix commutators and hence additional geometry. The underlying fields and dynamics must actually support those equations; changing the name of a two-charge model does not create four physical supercharges.

We therefore allow new bosons, fermions, boundary data and genuine interactions. The models studied below are supersymmetric quantum mechanics with complex coordinates and a holomorphic superpotential, often called Landau–Ginzburg models. Their fermions are represented by differential forms, and their Hamiltonian positivity follows from physical adjoints. A polynomial superpotential can create several isolated vacua and a nontrivial matrix of ground-state overlaps. A periodic complex coordinate introduces an additional global issue: some parameter insertions become multivalued, so ordinary chiral tt^* equations cannot automatically be applied without the periodic or flavor data [6].

Gaussianity is consequently a modeling choice, not the boundary between possible and impossible arithmetic constructions. Quadratic models are valuable controls because they can be solved exactly. An interacting theory is more useful only when its interaction changes the pairing or constraints relevant to (1.4). The calculations here give a concrete example of that distinction.

1.5 Overview of the results

First, replacing the real Morse coordinate by a complex cylinder produces an actual four-supercharge theory. It has a normalizable one-form vacuum for nonzero cylinder parameter. The gamma function

survives as a holomorphic integral, but this integral is not identified with the physical norm of that vacuum. Two added quadratic complex prime fields give both an Euler period and a physical metric proportional to $|1 - q|^{-2}$ in a specified holomorphic frame. The latter has zero local curvature, so its interpretation depends on the physical choice of source.

Second, a cross-coupled quartic prime theory has nine normalizable middle-degree vacua. At fixed nonzero quartic coupling, an exact differential equation excludes the Euler identity period on every flat rapid-decay cycle obeying the stated boundary conditions. This is an analytical exclusion of a class of theories and observables, not a failure to fit a numerical curve.

Third, scaling the quartic coupling with the square of the prime mass restores the Euler factor exactly. With $M = 1 - q$ and a fixed interaction shape $c > 0$, the identity period and physical identity-class metric have the forms

$$\mathcal{P}_c(M) = \frac{C_c}{M}, \quad h_c(M, \bar{M}) = \frac{H_c}{|M|^2}, \quad C_c, H_c > 0. \quad (1.7)$$

This shows that an interacting positive theory can carry the exact Euler dependence. The constants describe different quantities and are not known to satisfy $H_c = |C_c|^2$. Moreover, the mass dependence is generated by a field dilation: the mass deformation vanishes in the Jacobi ring, the algebra of holomorphic observables modulo the superpotential gradients. It therefore supplies no new ordinary chiral-curvature constraint along that parameter.

Finally, several apparently simple completions are excluded under explicit hypotheses. Finite ambient-holomorphic vacuum enlargement conflicts with the trace of the tt^* curvature equation. Scalar reversible ground transforms and scalar phase changes have the wrong off-diagonal signs for the full target. A positive square representing the pole correlations necessarily adds a diagonal energy, which must also be accounted for. Curvature equations alone leave a normalization freedom that changes the contact term. These restrictions still allow matrix or fermionic interference, nonlocal source maps and physically specified relative boundary pairings.

The resulting next problem is narrower than a search for an arbitrary complicated quantum theory. Starting with the exact interacting control (1.7), one should change the interaction shape so that its parameter operator is nonzero in the Jacobi ring, and identify the physical boundary pairing before attempting a large overlap calculation. A source-normalization-invariant comparison between its physical metric and squared period is one concrete diagnostic. The decisive advance would be a pairing identity that supplies the contact and poles together with the full input space. No such identity, all-support positivity theorem or Riemann-hypothesis proof is claimed here.

1.6 Organization and status

Section 2 derives the arithmetic starting identities. The subsequent sections construct the enlarged theory, analyze the interacting Euler sector, and develop the metric and completion restrictions. All numbered results specify their model and hypotheses. The mathematical and physical frameworks are attributed to external primary references; the particular algebraic tests and model comparisons are derived explicitly. This is an AI-assisted research draft. Internal cross-review and exact algebra checks do not replace independent specialist review or establish a priority claim.

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2 The arithmetic target and its positive starting models

Throughout, inner products are conjugate-linear in their first entry. Let $E_L : L^2(I_L) \rightarrow L^2(\mathbb{R})$ denote zero extension. The central Weil form is (1.1); no positivity assumption is made about its sum. For fixed L , all translated and pole terms are bounded forms. Since $B(\tau^2)$ grows logarithmically, its closed semibounded realization has form domain

$$\mathcal{D}_L = \left\{ f \in L^2(I_L) : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{E_L f}(\tau)|^2 d\tau < \infty \right\}. \quad (2.1)$$

The positive form K is closed by the Fourier multiplier construction, and bounded perturbations give the closed semibounded target operator W_L . Smooth compactly supported inputs are the initial arithmetic test domain. This operator formulation does not require selected test functions to be estimated individually, but an eventual factorization must hold on the full form domain.

2.1 Gamma kinetic energy as an exact positive response

The elementary partial-fraction expansion of the digamma difference gives

$$B(s) = \sum_{k \geq 0} \frac{2}{a_k} \frac{s}{s + a_k^2}, \quad a_k = 2k + \frac{1}{2}. \quad (2.2)$$

For example, the real part of $\psi(b + it) - \psi(b) = \sum_{k \geq 0} [(k + b)^{-1} - (k + b + it)^{-1}]$ is $\sum_{k \geq 0} t^2 / [(k + b)((k + b)^2 + t^2)]$; taking $b = 1/4$ and $t = \sqrt{s}/2$ proves (2.2).

For a mass $a > 0$, define

$$\mathcal{E}_a[F, u] = \frac{2}{a} \left(\|F - u\|_{L^2(\mathbb{R})}^2 + a^{-2} \|u'\|_{L^2(\mathbb{R})}^2 \right), \quad u \in H^1(\mathbb{R}).$$

Its unique minimizer solves $(1 - a^{-2} \partial_x^2)u = F$. In Fourier space,

$$\widehat{u}(\tau) = \frac{a^2}{a^2 + \tau^2} \widehat{F}(\tau), \quad \inf_u \mathcal{E}_a[F, u] = \frac{1}{2\pi} \int \frac{2}{a} \frac{\tau^2}{a^2 + \tau^2} |\widehat{F}(\tau)|^2 d\tau.$$

Summing these nonnegative channel energies and using monotone convergence gives

$$K[F] = \inf_{(u_k)} \sum_{k \geq 0} \mathcal{E}_{a_k}[F, u_k]. \quad (2.3)$$

The admissible fields have finite displayed energy. On (2.1), the channel minimizers achieve the sum; outside it the infimum is infinite. One must not split off the divergent unprepared sum $\sum_k (2/a_k) \|F\|^2$ before minimizing. Formula (2.3) is the exact positive bulk response used as the starting success.

The same kinetic form has a jump representation. Put

$$n_\gamma(r) = \frac{e^{-|r|/2}}{1 - e^{-2|r|}}, \quad r \neq 0.$$

Expanding this density as $\sum_{k \geq 0} e^{-a_k |r|}$ and using $\int_{\mathbb{R}} e^{-a|r|} (1 - \cos \tau r) dr = (2/a) \tau^2 / (a^2 + \tau^2)$ yields

$$K[F] = \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}} |F(x) - F(y)|^2 n_\gamma(x - y) dx dy. \quad (2.4)$$

The diagonal singularity is interpreted through the difference square.

2.2 An exact finite-prime ground-state amplitude

Let $\mathcal{S}$ be finite and $\sigma > 0$. On

$$\mathcal{H}_{\mathcal{S}} = L^2(\mathbb{R}, dy) \otimes \bigotimes_{p \in \mathcal{S}} \ell^2(\mathbb{N}_0),$$

use the closed Morse operator and bounded prime differences

$$A_{\sigma/2} = \partial_y + \frac{e^y - \sigma/2}{2}, \quad D_p = T - p^{-\sigma/2} I, \quad (T\xi)_n = \xi_{n+1}.$$

Each defines a two-term complex with its physical Hilbert adjoint; their graded tensor product has a positive supersymmetric Hamiltonian. The Morse kernel is obtained by integrating a first-order equation, while $D_p \xi = 0$ gives a geometric sequence. The adjoint kernels vanish: the Morse adjoint solution grows at $+\infty$, and the zeroth component of $D_p^* \xi = 0$ forces every component to vanish. Consequently the product has a unique bosonic vacuum ray,

$$\Psi_{\sigma}(y, \mathbf{n}) = \pi^{-\sigma/4} \exp\left(\frac{\sigma y/2 - e^y}{2}\right) \prod_{p \in \mathcal{S}} p^{-\sigma n_p/2}. \quad (2.5)$$

Morse supersymmetry and zeta-related Morse Fourier amplitudes have existing antecedents [2, 3]; no priority is claimed for these elementary factor constructions.

The multiplication operator

$$X = \frac{y}{2} - \sum_{p \in \mathcal{S}} n_p \log p - \frac{\log \pi}{2}$$

is self-adjoint on its maximal multiplication domain. Substituting $t = e^y$ and summing geometric series proves

$$\|\Psi_{\sigma}\|^2 = \Lambda_{\mathcal{S}}(\sigma), \quad \langle \Psi_{\sigma}, e^{i\tau X} \Psi_{\sigma} \rangle = \Lambda_{\mathcal{S}}(\sigma + i\tau). \quad (2.6)$$

The state normalization in (2.5) is specified data. The Hamiltonian only determines the ray.

2.3 A positive derivative and the missing contact

The product $\Lambda_{\mathcal{S}}$ is nonzero on $\operatorname{Re} z > 0$. Its logarithmic derivative is

$$\frac{\Lambda'_{\mathcal{S}}(z)}{\Lambda_{\mathcal{S}}(z)} = -\frac{\log \pi}{2} + \frac{1}{2} \psi(z/2) - \sum_{p \in \mathcal{S}} \frac{\log p}{p^z - 1}. \quad (2.7)$$

Taking the real part and subtracting its value on the real axis gives

$$\begin{aligned} \partial_{\sigma} \log \frac{|\Lambda_{\mathcal{S}}(\sigma + i\tau)|^2}{\Lambda_{\mathcal{S}}(\sigma)^2} \Big|_{\sigma=1/2} &= B(\tau^2) \\ &+ 2 \sum_{p \in \mathcal{S}, m \geq 1} (\log p) p^{-m/2} [1 - \cos(m \log p \tau)]. \end{aligned} \quad (2.8)$$

Every term on the right is nonnegative. Define $J_{\mathcal{S},L}$ by compressing this Fourier multiplier through zero extension. If $a_{p,m} = (\log p) p^{-m/2}$, then

$$J_{\mathcal{S},L}[f] = K[E_L f] + \sum_{p \in \mathcal{S}, m \geq 1} a_{p,m} \|E_L f - U_{m \log p} E_L f\|^2.$$

The repetitions are summable for finite $\mathcal{S}$.

Proposition 2.1 (Exact contact decomposition). *Suppose $\mathcal{S}$ contains every prime p with $\log p < L$. Put*

$$\kappa_{\mathcal{S}} = -w_0 + 2 \sum_{p \in \mathcal{S}, m \geq 1} a_{p,m} > 0, \quad P_L[f] = 2|C(f)|^2 - 2|S(f)|^2.$$

On $\mathcal{D}_L$,

$$Q_L[f] = J_{\mathcal{S},L}[f] - \kappa_{\mathcal{S}} \|f\|^2 + P_L[f]. \quad (2.9)$$

Proof. Expand each translated difference square as $2\|F\|^2 - 2\operatorname{Re}\langle F, U_d F \rangle$. If $d \geq L$, the correlation vanishes by disjoint support. The surviving correlations are exactly the prime terms in (1.1); the sum of all the diagonal contributions is cancelled by the definition of $\kappa_{\mathcal{S}}$. $\square$

Additional inactive primes change J and κ by cancelling diagonal quantities. Their inclusion is harmless only when that cancellation is kept. An independent all-prime jump sum is divergent on every nonzero compactly supported input: for sufficiently large p , its first translation has disjoint support and contributes $2(\log p)p^{-1/2}\|f\|^2$. The divergence follows already from the divergence of $\sum_p 1/p$. Thus the finite-prime control is not a justification for an unrestricted positive jump limit.

The normalized squared amplitude in (2.8) is a positive-definite function before differentiation, but its logarithmic derivative is a different observable. Equation (2.9), rather than the positivity of the characteristic function, is what specifies the missing arithmetic problem.

2.4 The vacuum metric is not the squared amplitude

A chosen complex extension of (2.5) is

$$\Psi_z = e^{zX/2} e^{-e^y/2}, \quad h(z, \bar{z}) = \|\Psi_z\|^2 = \Lambda_{\mathcal{S}}(\operatorname{Re} z).$$

With $\sigma = \operatorname{Re} z$, the second derivative of $\log \Lambda_{\mathcal{S}}(\sigma)$ is the variance of X in the normalized state, hence strictly positive. In a holomorphic line frame,

$$\partial_{\bar{z}} \partial_z \log h = \frac{1}{4} (\log \Lambda_{\mathcal{S}})''(\sigma) > 0.$$

In contrast, $|\Lambda_{\mathcal{S}}(z)|^2$ has zero logarithmic curvature wherever $\Lambda_{\mathcal{S}}$ is holomorphic and nonzero. Confusing these two metrics erases precisely the geometric obstruction motivating the enlargement.

3 A positive four-supercharge enlargement

An enlargement of the field space provides a concrete way to obtain four real supercharges without retaining the spectrum of a two-supercharge model. The construction below adds an angular coordinate to the real Morse variable. It retains the gamma function as a holomorphic period, but its physical vacuum is a differential form on the enlarged target. Adding two complex oscillators then produces a physical Euler-factor metric in an explicitly specified parameter frame. These statements concern positive quantum theories and identifiable observables; they do not identify their vacuum pairing with the complete Weil form.

3.1 The cylinder and its supersymmetry algebra

Let

$$\mathcal{X} = \mathbb{R} \times (\mathbb{R}/2\pi\mathbb{Z}), \quad Y = y + i\theta, \quad \mathcal{H} = L^2\Omega^*(\mathcal{X}; \mathbb{C}), \quad (3.1)$$

with flat metric $dy^2 + d\theta^2$ and the ordinary positive inner product on complex differential forms. Write E_y, E_{θ} for exterior multiplication by $dy, d\theta$, and I_y, I_{θ} for their Hilbert adjoints. Thus

$\{E_i, I_j\} = \delta_{ij}$, while the anticommutators of two exterior multiplications or two contractions vanish. Initially take $a > 0$ and define

$$\mathcal{W}_a(Y) = \frac{1}{2}(e^Y - aY), \quad h = \text{Re } \mathcal{W}_a = \frac{1}{2}(e^y \cos \theta - ay). \quad (3.2)$$

Although $\mathcal{W}_a$ has an additive period, h is single valued for real a . In particular $\Delta h = h_{yy} + h_{\theta\theta} = 0$.

On smooth compactly supported periodic forms define

$$\begin{aligned} Q_1 &= E_y(\partial_y + h_y) + E_\theta(\partial_\theta + h_\theta), \\ Q_1^\dagger &= I_y(-\partial_y + h_y) + I_\theta(-\partial_\theta + h_\theta), \\ Q_2 &= -E_y(\partial_\theta - h_\theta) + E_\theta(\partial_y - h_y), \\ Q_2^\dagger &= I_y(\partial_\theta + h_\theta) + I_\theta(-\partial_y - h_y). \end{aligned} \quad (3.3)$$

Equivalently, $Q_1 = d + dh \wedge$ and $Q_2 = d^c - d^c h \wedge$, where $d^c = E_\theta \partial_y - E_y \partial_\theta$. All adjoints in (3.3) refer to (3.1); an indefinite fermionic pairing is not involved.

For a general smooth real h , the exterior relations give

$$\begin{aligned} Q_1^2 &= Q_2^2 = 0, \quad \{Q_1, Q_2^\dagger\} = 0, \\ \{Q_1, Q_2\} &= -2(\Delta h)E_y E_\theta, \\ \{Q_1, Q_1^\dagger\} - \{Q_2, Q_2^\dagger\} &= 2(\Delta h)(N - 1), \quad N = E_y I_y + E_\theta I_\theta. \end{aligned} \quad (3.4)$$

For example, the coefficient of $E_y E_\theta$ in the mixed nilpotent anticommutator is $[\partial_y + h_y, \partial_y - h_y] + [\partial_\theta + h_\theta, \partial_\theta - h_\theta] = -2\Delta h$. The remaining identities follow by the same exterior expansion. Harmonicity therefore gives a common Hamiltonian $H = \{Q_1, Q_1^\dagger\} = \{Q_2, Q_2^\dagger\}$ and four real charges

$$\begin{aligned} R_1 &= Q_1 + Q_1^\dagger, & R_2 &= i(Q_1 - Q_1^\dagger), & \{R_i, R_j\} &= 2\delta_{ij}H. \\ R_3 &= Q_2 + Q_2^\dagger, & R_4 &= i(Q_2 - Q_2^\dagger), \end{aligned} \quad (3.5)$$

Their common square is

$$\begin{aligned} H &= -\partial_y^2 - \partial_\theta^2 + \frac{1}{4}|e^Y - a|^2 \\ &\quad + e^y [\cos \theta (E_y I_y - E_\theta I_\theta) - \sin \theta (E_y I_\theta + E_\theta I_y)]. \end{aligned} \quad (3.6)$$

This is an exponential bosonic interaction with a Hessian coupling to the fermions. In the corresponding normal-ordered coherent-state description its Euclidean action is

$$S_E = \int dt \left\{ \frac{1}{4}(\dot{y}^2 + \dot{\theta}^2) + |\nabla h|^2 + \bar{c}_i \dot{c}_i + 2\bar{c}_i h_{ij} \dot{c}_j \right\}. \quad (3.7)$$

The normalization follows from the kinetic operator $-\Delta$ in (3.6); the scalar normal-ordering contribution $-\Delta h$ vanishes. Positivity is the operator statement $H \geq 0$, not positivity of every fermionic path-integral integrand.

3.2 Closed operators and the physical vacuum

The displayed algebra has a closed realization. Each R_i is a Dirac-type first-order operator with smooth Hermitian multiplication potential on a complete flat manifold. Choose compact-support cutoffs χ_L tending to one with $|d\chi_L| \leq C/L$. The potential commutes with the cutoff,

so $\|[R_i, \chi_L]\| \leq C/L$. A deficiency vector satisfying $R_i^* \psi = \pm i\psi$ is locally regular by ellipticity. Integration by parts with $\chi_L^2 \psi$, followed by taking the imaginary part, gives

$$\|\chi_L \psi\|^2 \leq \frac{C}{L} \|\chi_L \psi\| \|\psi\|. \quad (3.8)$$

Consequently both deficiency spaces vanish. This proves essential self-adjointness on the stated common core. The same argument applies to every nonzero real linear combination of the R_i .

On that core the graph norms of the four charges coincide, by (3.5). Their closures therefore have a common quadratic-form domain, and their squares define the same nonnegative self-adjoint H . Polarizing the identities for real linear combinations proves the mixed algebra as a form identity. Since each closed R_i commutes with the spectral resolution of $R_i^2 = H$, all products in (3.5) are defined on $\text{Dom } H$, where the operator identities follow. This argument also applies to the product targets used below. It is compatible with the noncompact Witten-operator framework of [7]; the cutoff argument itself does not require the vacuum-counting theorem.

Restricting the function h to $\theta = 0$ gives $(e^y - ay)/2$, but restricting a function is not a quantum reduction. For example, the scalar real-line operator obtained from $A = \partial_y + (e^y - a)/2$ is

$$A^\dagger A = -\partial_y^2 + \frac{1}{4}(e^y - a)^2 - \frac{1}{2}e^y. \quad (3.9)$$

The scalar cylinder Hamiltonian has no negative Hessian term because $\Delta h = 0$. In degree one its Hessian term has eigenvalues $\pm e^y$, rather than $-e^y/2$. A delta function at a fixed angle is not a state of (3.1). Moreover, e^{-h} grows exponentially near $\theta = \pi$ as $y \rightarrow +\infty$, and e^h grows near $\theta = 0$. Neither supplies a normalizable scalar or top-form vacuum.

Proposition 3.1. *For every $a \in \mathbb{C} \setminus \{0\}$, the cylinder Hamiltonian with periodic differential forms has exactly one complex normalizable zero-energy state. It is a one-form. The assertion uses the well-tame Morse inequalities of [7] at large Witten deformation, together with the continuation argument below; it does not compute a vacuum norm in an arithmetic source frame.*

Proof. First let $a > 0$ and replace h by Th , $T > 0$. There is exactly one critical point, $(y, \theta) = (\log a, 0)$, of Morse index one. The complete flat cylinder has bounded geometry, and

$$\liminf_{|y| \rightarrow \infty} |dh| = \frac{a}{2} > 0, \quad \frac{|\text{Hess } h|}{|dh|^2} \rightarrow 0. \quad (3.10)$$

At the negative end the Hessian tends to zero and $|dh| \rightarrow a/2$; at the positive end $|dh|$ grows like $e^y/2$ while the Hessian grows like e^y . These are the well-tame hypotheses in [7]. Their strong Morse inequalities, including equality of the full alternating sums, give graded Dirac index -1 for sufficiently large T . This step uses the analytic theorem, rather than an assumed identification of critical points with exact quantum states. It does not require the additional Thom–Smale comparison theorem.

The graded Dirac operator remains Fredholm for all $T > 0$. At $y \rightarrow -\infty$ its square tends to $-\partial_y^2 - \partial_\theta^2 + T^2 a^2/4$; at the other end its potential tends to $+\infty$, since $T^2 |dh|^2$ dominates the Hessian. These estimates are uniform on compact positive T intervals and give a positive lower bound outside a compact set. Local elliptic compactness then gives Fredholmness. The estimate $|\text{Hess } h| \leq \varepsilon |dh|^2 + C_\varepsilon$ also makes the graph domains equivalent on those intervals, so the family is continuous in graph norm and its index is constant. This is the required continuation from the large-deformation Morse calculation [2, 7].

In degrees zero and two the squared operator is simply $-\Delta + T^2 |dh|^2$. A zero state would have both vanishing weak gradient and vanishing product with $|dh|$, and hence must be zero. Thus the index -1 implies precisely one degree-one vacuum.

For complex a , the globally defined one-form dh is described explicitly below. Its variation with a is a bounded constant one-form. The negative-end gap becomes $T^2|a|^2/4$ and the positive-end estimate persists. Any path in $\mathbb{C} \setminus \{0\}$ therefore gives a Fredholm continuation from positive real a . The even-degree kernels remain zero because $\Delta h = 0$ locally. The same count follows at $T = 1$. At $a = 0$ the end gap closes, so this argument makes no assertion there. $\square$

Writing the vacuum as $\omega = P dy + Q d\theta$, its equations are

$$\begin{aligned} \partial_y Q - \partial_\theta P + h_y Q - h_\theta P &= 0, \\ -\partial_y P - \partial_\theta Q + h_y P + h_\theta Q &= 0. \end{aligned} \tag{3.11}$$

These determine the physical state on the full cylinder. No scalar gamma wavefunction is substituted for their solution.

3.3 Periodic parameters and the gamma period

For $a = a_R + ia_I$ one has locally

$$h = \frac{1}{2}(e^y \cos \theta - a_R y + a_I \theta), \quad h(y, \theta + 2\pi) - h(y, \theta) = \pi a_I. \tag{3.12}$$

Although h need not be globally a function, its differential is a smooth closed real one-form on the cylinder. The supercharges use this one-form and its complex-structure rotation, so remain globally defined on periodic forms. Their definition is not globally a conjugation by a single-valued e^h . Similarly, $\mathcal{W}_a(Y + 2\pi i) - \mathcal{W}_a(Y) = -\pi i a$, and the proposed chiral insertion $\partial_a \mathcal{W}_a = -Y/2$ is multivalued. The local holomorphic dependence of a superpotential thus does not by itself supply an ordinary globally defined rank-one chiral tt^* problem. Periodic Landau–Ginzburg theories with deck and flavor data provide a relevant framework [6]; any such extra data must be included explicitly in a proposed arithmetic pairing.

The ordinary Berry connection of the specific one-vacuum realization above is locally flat. Indeed, H has real coefficients for every complex a , and its isolated one-dimensional kernel admits a real normalized generator ψ locally. Reality and differentiation of $\langle \psi, \psi \rangle = 1$ imply $\langle \psi, d\psi \rangle = 0$. A possible global sign holonomy does not change this local conclusion. Parameter connections with additional insertions or flavor structure are separate constructions; the distinction matters when using supersymmetric Berry equations [5].

There is nevertheless an exact holomorphic gamma period:

$$\mathcal{P}_\gamma(a) = \int_{Y \in \mathbb{R}} e^{-2\mathcal{W}_a(Y)} dY = \int_0^\infty t^{a-1} e^{-t} dt = \Gamma(a), \quad \operatorname{Re} a > 0. \tag{3.13}$$

The factor two in the exponent is part of the definition. For complex a the real path is a convergent relative cycle, not necessarily a steepest-descent thimble. Such periods belong to the holomorphic boundary quantities of Landau–Ginzburg theory [6]. Equation (3.13) is not an evaluation of the Hermitian norm of the physical one-form in (3.11).

3.4 Two complex oscillators and a physical Euler metric

Introduce independent parameters a, q with $\operatorname{Re} a > 0$, $|q| < 1$, put $M = 1 - q$, and enlarge the target to $\mathcal{X} \times \mathbb{C}^2$ with its product flat metric. The superpotential is

$$\mathcal{W}_{a,q}(Y, U, V) = \frac{1}{2} \left[e^Y - aY + \frac{M}{2}(U^2 + V^2) \right]. \tag{3.14}$$

This introduces two further complex bosons and their form-valued fermions. On a flat Kähler target the preceding construction extends with d^c defined by the complex structure. The required Hessian condition is pluriharmonicity, rather than merely a vanishing trace; it holds for $h = \operatorname{Re} \mathcal{W}_{a,q}$. Alternatively, the graded tensor product of the three two-dimensional complexes verifies the four-charge algebra directly. The common-core closure argument above applies on this complete product target. In particular,

$$|\nabla h|^2 = \frac{1}{4}|e^Y - a|^2 + \frac{|M|^2}{4}(|U|^2 + |V|^2). \quad (3.15)$$

Since $\operatorname{Re} M > 0$, elementary integration on the real cycle gives

$$\frac{1}{2\pi} \int_{\mathbb{R}^3} e^{-2\mathcal{W}_{a,q}} dY dU dV = \frac{\Gamma(a)}{1-q}. \quad (3.16)$$

For a fixed prime p , setting $a = z/2$, $q = p^{-z}$ and specifying the source multiplier $\pi^{-z/2}$ produces $\pi^{-z/2}\Gamma(z/2)(1-p^{-z})^{-1}$ for $\operatorname{Re} z > 0$. Neither this restriction of parameters nor its Fourier variable $\operatorname{Im} z$ identifies an arithmetic coordinate with physical Euclidean time.

The oscillator vacuum metric is separately computable. For one field with $\mathcal{W}_U = MU^2/4$, write $M = re^{i\phi}$ and $U' = e^{i\phi/2}U = x' + iy'$. Then

$$h_U = \frac{r}{4}(x'^2 - y'^2), \quad \psi_M = e^{-r|U|^2/4} dy', \quad \|\psi_M\|^2 = \frac{2\pi}{r}. \quad (3.17)$$

The form ψ_M satisfies $Q_1\psi_M = Q_1^\dagger\psi_M = 0$. Equivalently, the scalar oscillator ground energy is r , cancelled by the occupied negative Hessian direction, whose contribution is $-r$. Every other oscillator state has positive energy, proving uniqueness. Consequently the product theory also has a unique physical vacuum for $aM \neq 0$.

To make the parameter frame explicit, define

$$\begin{aligned} s_M &= (2\pi)^{-1/2} e^{-i\phi/2} \psi_M \\ &= \frac{e^{-r|U|^2/4}}{2i\sqrt{2\pi}} \left(dU - \frac{\bar{M}}{|M|} d\bar{U} \right). \end{aligned} \quad (3.18)$$

The second expression is single valued for $M \neq 0$. If P_M denotes the physical ground-state projector, then $P_M \partial_{\bar{M}} s_M = 0$. To verify this without identifying different Hodge realizations, normalize the real state as $\chi_M = \sqrt{r/(2\pi)} \psi_M$. It has zero local Berry connection, and $s_M = M^{-1/2} \chi_M$ on a local branch. The coefficient is holomorphic, which proves the assertion directly in the Hilbert space of real differential forms complexified. Thus

$$\langle s_M, s_M \rangle = |M|^{-1}, \quad g_{UV} = |M|^{-2} = |1 - q|^{-2}. \quad (3.19)$$

No identification with a twisted-Dolbeault identity class is needed for this explicit calculation.

The coefficient in (3.18) fixes the source normalization. In the associated tensor frame the full physical metric is $g_\gamma(a, \bar{a})|1 - q|^{-2}$, where the cylinder factor has not been evaluated. It is distinct from the period (3.16). Furthermore, $\partial_M \partial_{\bar{M}} \log g_{UV} = 0$ away from zero, and multiplication of the two-field frame by M gives unit metric. Thus the denominator has meaning relative to the specified source; positivity alone does not select it. At $M = 0$ oscillator confinement and its normalizable vacuum disappear. These facts provide a positive physical benchmark, while leaving the arithmetic contact and full pole pairing undetermined.

3.5 An affine gamma–prime coupling fails two tests

A direct attempt to couple the sectors replaces their constant mass by $M + \lambda e^Y$:

$$\mathcal{W}_{a,q}^\lambda = \frac{1}{2} \left[e^Y - aY + \frac{M + \lambda e^Y}{2} (U^2 + V^2) \right]. \quad (3.20)$$

For real $a, M, \lambda > 0$ its real-cycle integral converges, and integrating U, V gives

$$\begin{aligned} \mathcal{P}_\lambda(a, M) &= \int_0^\infty \frac{t^{a-1} e^{-t}}{M + \lambda t} dt < \frac{\Gamma(a)}{M}, \\ \frac{d}{d\lambda} \mathcal{P}_\lambda \Big|_{0+} &= -\frac{\Gamma(a+1)}{M^2}. \end{aligned} \quad (3.21)$$

The inequality and the right derivative are exact analytic checks of the amplitude. They are not numerical positivity bounds.

There is a separate obstruction in the full complex target. At $t_0 = -M/\lambda \neq 0$, the critical equations admit

$$e^Y = t_0, \quad U^2 + V^2 = \frac{2(a - t_0)}{\lambda t_0}. \quad (3.22)$$

This is a noncompact complex critical quadric: writing $(U + iV)(U - iV)$ as the displayed constant exhibits points escaping to infinity. Thus a convergent real contour conceals nonisolated classical vacua elsewhere in the physical field space. The isolated-vacuum hypotheses of a standard massive ground-state construction cannot simply be assumed. This observation does not by itself prove a vanishing quantum spectral gap.

The period obstruction has a useful scoped extension. Fix q and suppose a mass $m(t, q)$ is nonzero on $t > 0$ and satisfies the integrability assumptions for uniqueness of its Mellin transform. If

$$\int_0^\infty t^{a-1} e^{-t} m(t, q)^{-1} dt = \frac{\Gamma(a)}{1 - q} \quad (3.23)$$

holds throughout an open Mellin strip for independently variable a , then $m(t, q) = 1 - q$ almost everywhere on that contour. Indeed, subtract the two integrands, fix a real part $\operatorname{Re} a = b$ in the strip, and put $t = e^x$. The resulting L^1 function of x has identically zero Fourier transform, so vanishes almost everywhere. This excludes a nonconstant quadratic mass preserving the entire factorized period within this class. It does not exclude a match only on the linked locus $q = p^{-2a}$, a different insertion or boundary problem, or a nonquadratic prime theory. Those alternatives require additional structure beyond the positive product Hamiltonian constructed here.

4 Interacting prime fields: an exclusion and an exact Euler metric

The quadratic prime fields give an exact control, but only one vacuum. We now replace them by the cross-coupled polynomial

$$F_{M,g,\lambda}(U, V) = \frac{M}{2}(U^2 + V^2) + \frac{g}{4}(U^4 + V^4) + \frac{\lambda}{2}U^2V^2. \quad (4.1)$$

The prime superpotential is $\mathcal{W} = F/2$; adjoining the cylinder gives $\mathcal{W}_{\text{full}} = \frac{1}{2}[e^Y - aY + F]$. All physical target metrics in this section are flat. The factor $1/2$ matters: the holomorphic period considered below has weight $e^{-2\mathcal{W}} = e^{-F}$, while the physical vacuum metric is defined by the supersymmetric Hilbert problem.

4.1 Nine isolated physical vacua

For definiteness take $M \neq 0$, $g > 0$ and $\lambda = g/2$. The critical equations are

$$U(M + gU^2 + \lambda V^2) = 0, \quad V(M + gV^2 + \lambda U^2) = 0.$$

There is one origin, four axis points with the nonzero coordinate squared equal to $-M/g$, and four points with

$$U^2 = V^2 = -M/(g + \lambda).$$

The Hessian determinant is M^2 at the origin, $-2M^2(1 - \lambda/g)$ at an axis point, and $4(g^2 - \lambda^2)U^2V^2$ at a point with both coordinates nonzero. All nine points are nondegenerate.

The homogeneous cubic gradient of the quartic part has no zero on the complex unit sphere when $g \neq 0$ and $g^2 \neq \lambda^2$. Compactness of that sphere gives constants with

$$|\nabla F| \geq c_1 R^3 - c_2 R, \quad |\text{Hess } F| \leq c_3(1 + R^2).$$

The gradient square therefore dominates every fixed multiple of the Hessian at infinity. This verifies strong tameness on the complete Kähler–Stein target $\mathbb{C}^2$. The corresponding Landau–Ginzburg Hodge theorem identifies the physical vacuum space with the Jacobi classes in middle degree [8]; its dimension here is nine. This use of the theorem supplies existence and the cohomological identification, not a formula for the metric.

At fixed g, λ , the superpotential-variation observable is multiplication by $[\partial_M \mathcal{W}] = [(U^2 + V^2)/4]$. Its critical-point values are 0, $-M/(4g)$ and $-M/[2(g + \lambda)]$, with multiplicities 1, 4, 4. In the twisted-Dolbeault convention specified below, the Higgs operator is half this variation. It is still nonscalar, so this interacting theory has room for nontrivial matrix geometry. Whether its observable retains the arithmetic factor is a separate test.

4.2 Every admissible identity period obeys a differential equation

Let Γ_M be a locally flat family of rapid-decay relative cycles, with no finite-field boundary and sufficient decay for the following polynomial moments and integrations by parts. The coefficients of a transported cycle are fixed in relative homology; arbitrary constant complex combinations are allowed. Parameter-dependent fitted coefficients, singular insertions and nonvanishing boundary terms are not part of this class. The usual thimble description is one realization of these assumptions [10].

Write

$$\langle P \rangle_M = \frac{1}{2\pi} \int_{\Gamma_M} P(U, V) e^{-F} dU \wedge dV, \quad s = U^2 + V^2, \quad t = U^2 V^2,$$

and set $I = \langle 1 \rangle_M$, $A = \langle s \rangle_M$, $B = \langle t \rangle_M$. Differentiation keeps g, λ fixed and gives

$$I' = -A/2, \quad A' = -\langle s^2 \rangle_M/2, \quad B' = -\langle st \rangle_M/2.$$

The exact two-forms obtained by differentiating Ue^{-F} and Ve^{-F} , and then UV^2e^{-F} and VU^2e^{-F} , have vanishing boundary integrals. Their expansions are

$$2I - MA - g\langle U^4 + V^4 \rangle_M - 2\lambda B = 0, \quad A - 2MB - (g + \lambda)\langle st \rangle_M = 0.$$

Consequently

$$\begin{aligned} I' &= -A/2, \\ A' &= -I/g + MA/(2g) - (g - \lambda)B/g, \\ B' &= -A/[2(g + \lambda)] + MB/(g + \lambda). \end{aligned} \tag{4.2}$$

For $g \neq \lambda$ one may eliminate the moments using

$$A = -2I', \quad B = \frac{2gI'' - MI' - I}{g - \lambda}.$$

Substitution yields

$$2g(g + \lambda)I''' - (3g + \lambda)MI'' + [M^2 - (3g + \lambda)]I' + MI = 0.$$

Only the subsystem probed by the identity insertion and mass derivatives has rank at most three; this does not reduce the full nine-dimensional vacuum space to three dimensions.

Theorem 4.1 (Fixed-coupling period exclusion). *Fix $g > 0$ and $\lambda = g/2$. Every identity period in the preceding cycle class obeys*

$$\mathcal{L}_g I := 6g^2 I''' - 7gMI'' + (2M^2 - 7g)I' + 2MI = 0. \quad (4.3)$$

No such period can equal a nonzero constant multiple of $1/M$ on a nonempty open parameter region.

Proof. The equation is the specialization of (4.2). Direct substitution gives

$$\mathcal{L}_g(1/M) = -\frac{g(7M^2 + 36g)}{M^4}, \quad (4.4)$$

which is not identically zero on any open region. Every constant combination of admissible periods obeys the same linear equation, so cancellation among saddle contributions cannot change the conclusion. $\square$

The coefficients of (4.3) are entire in M and its leading coefficient is nonzero. Hence every local solution continues to an entire solution in the finite M -plane. Fixed quartic confinement persists at $M = 0$ even though critical points collide; the identity period cannot acquire the Gaussian Euler pole there. Stokes changes of an asymptotic cycle basis do not create a pole in a transported solution. This is stronger than the strict suppression of the real-cycle integral, which follows immediately from the positive real quartic term.

4.3 A mass-dependent interaction restores the period exactly

The fixed-coupling hypothesis is substantive. Let $c > 0$ be constant and choose

$$g(M) = cM^2, \quad \lambda(M) = cM^2/2. \quad (4.5)$$

With

$$G_c(A, B) = \frac{1}{2}(A^2 + B^2) + \frac{c}{4}(A^4 + B^4 + A^2B^2),$$

one has the holomorphic change-of-variables identity

$$F_M(U, V) = G_c(\sqrt{M}U, \sqrt{M}V). \quad (4.6)$$

For positive real M use the real cycle. For nonzero complex M use its dilation transport on a local square-root branch. Then

$$I_c(M) = \frac{C_c}{M}, \quad C_c = \frac{1}{2\pi} \int_{\mathbb{R}^2} e^{-G_c(A, B)} dA dB > 0. \quad (4.7)$$

The equality follows from $dA \wedge dB = M dU \wedge dV$, with no parameter-dependent fitted cycle coefficients. The fixed-coupling differential equation does not apply, because total differentiation now includes $g'(M)$ and $\lambda'(M)$.

For each $M \neq 0$, the theory remains strongly tame and has nine nondegenerate critical points. At $M = 0$, all coefficients in F_M vanish and confinement is lost. The eight nonzero critical points move to infinity like $M^{-1/2}$. The degeneration gives a physical location for the recovered pole.

4.4 The interacting physical metric has the same mass dependence

We specify the physical frame by the identity holomorphic volume class. The strongly tame Landau–Ginzburg Hodge identification gives its middle-degree harmonic representative, using the fixed normalization $\mathcal{W} = F/2$. The bridge to the twisted Dolbeault realization is explicit in the charge conventions of Section 3:

$$D = \frac{Q_1 - iQ_2}{2} = \bar{\partial} + \frac{1}{2}\partial\mathcal{W}\wedge, \quad \{D, D^\dagger\} = H/2. \quad (4.8)$$

Thus it has the same physical harmonic kernel. Fan’s differential $\bar{\partial} + \partial\mathfrak{f}\wedge$ corresponds here to $\mathfrak{f} = \mathcal{W}/2 = F/4$. Its canonically normalized Higgs operator is $[\partial_M\mathfrak{f}] = \frac{1}{2}[\partial_M\mathcal{W}]$. This constant scaling does not change the Jacobi ideal, but it fixes the normalization of any future nonzero curvature equation. All comparisons below use this same physical realization; a scalar period is not declared to be a norm.

Proposition 4.2 (Interacting physical Euler dependence). *Let α_c be the physical harmonic representative of the identity class $dA \wedge dB$ for the reference theory $G_c/2$, and put $H_c = \|\alpha_c\|^2 > 0$. For the family (4.6), the identity-class metric in the (U, V) volume frame is*

$$h_c(M, \bar{M}) = \frac{H_c}{|M|^2}, \quad M \neq 0. \quad (4.9)$$

Proof. Let $\phi_M(U, V) = (\sqrt{M}U, \sqrt{M}V)$. It pulls back the reference flat metric to $|M|$ times the flat metric and pulls back the superpotential to the one at M . For a constant metric rescaling $g \mapsto ag$, a degree-one twisted exterior differential has adjoint and Laplacian

$$d_h^{\dagger, ag} = a^{-1}d_h^{\dagger, g}, \quad \Delta_{h, ag} = a^{-1}\Delta_{h, g}.$$

The same degree-counting statement holds in the twisted Dolbeault realization of the Jacobi identification. Thus constant rescaling changes no harmonic kernel. Pullback is consequently compatible with the harmonic representative and its holomorphic volume class.

In real dimension four, the norm of a degree-two form satisfies $\|\beta\|_{ag}^2 = a^{4/2-2}\|\beta\|_g^2 = \|\beta\|_g^2$. It follows that ϕ_M^* is an isometry on the middle-degree harmonic spaces when both original theories use the flat metric. Since $\phi_M^*(dA \wedge dB) = M dU \wedge dV$, uniqueness and linearity of harmonic representatives give

$$\alpha_M = M^{-1}\phi_M^*\alpha_c,$$

which proves (4.9). Reversing the square-root branch composes with $(A, B) \mapsto (-A, -B)$. This preserves the potential, metric and identity volume class, so this source is unchanged; other classes may have monodromy. $\square$

For $M = 1 - q$, division by the fixed reference norm gives exactly $|1 - q|^{-2}$. This is a physical interacting Euler metric. It does not identify H_c with $|C_c|^2$. A single source rescaling changes a period linearly and a norm quadratically, and therefore cannot independently set both constants to one. The normalized period and metric ratios at $M = 1$ are exact without that extra assertion.

4.5 Why this interaction does not yet constrain the contact

Differentiate (4.6) along the family (4.5). The chain rule gives

$$\partial_M F_M = \frac{U\partial_U F_M + V\partial_V F_M}{2M}. \quad (4.10)$$

The right side belongs to the Jacobian ideal. Thus the projected chiral mass operator is zero in $\mathbb{C}[U, V]/(\partial_U F_M, \partial_V F_M)$. In an ordinary chiral tt^* description this makes the curvature in the M direction zero. Correspondingly,

$$\partial_M \partial_{\bar{M}} \log h_c = 0 \quad (M \neq 0),$$

and the holomorphic frame change $\alpha_M \mapsto M \alpha_M$ removes the local denominator. The interaction is genuine at fixed c , but varying M only dilates it. The source frame and the singular degeneration carry the Euler dependence.

This establishes both a positive result and a criterion for the next theory. Interactions can preserve an exact arithmetic local factor; they need not destroy it as a matter of principle. To improve the missing pairing, however, one needs a parameter deformation or boundary constraint that contributes more than the field dilation already accounted for in (4.10).

5 Finite-rank metric constraints and a semisimple control

Additional vacua can compensate the curvature of an arithmetic ground-state line without introducing negative norms. They do not, however, automatically preserve its physical state family or its arithmetic observable. We establish two finite-rank restrictions and then give a positive rank-two solution of the metric equations that illustrates the distinction.

5.1 The metric equation and its determinant

Let $z = \sigma + i\eta$ be an ordinary chiral parameter. In an intrinsic holomorphic frame, write the positive Hermitian vacuum metric as g and set

$$A_z = g^{-1} \partial_z g, \quad \mathcal{K} = \partial_{\bar{z}}(g^{-1} \partial_z g), \quad C^{\dagger g} = g^{-1} C^* g. \quad (5.1)$$

Here C^* denotes conjugate transpose. Our convention for the ordinary chiral tt^* equation is

$$\partial_{\bar{z}} C = 0, \quad \mathcal{K} = [C, C^{\dagger g}]. \quad (5.2)$$

Indeed $[D_z, D_{\bar{z}}] = -\mathcal{K}$ for $D_z = \partial_z + A_z$, $D_{\bar{z}} = \partial_{\bar{z}}$, and (5.2) gives flatness of $D_z + \zeta^{-1} C$ and $D_{\bar{z}} + \zeta C^{\dagger g}$. The vacuum-metric interpretation originates in topological–antitopological fusion [4]; the parameter-multiplet hypothesis is essential to the supersymmetric Berry-connection derivation [5]. The existence of four formal supercharges alone does not identify a specified parameter family with (5.2).

For a finite set of primes S , the coherent-state benchmark has

$$h(\sigma) = \Lambda_S(\sigma), \quad \Lambda_S(z) = \pi^{-z/2} \Gamma(z/2) \prod_{p \in S} (1 - p^{-z})^{-1}, \quad \operatorname{Re} z > 0. \quad (5.3)$$

Writing $u = \log h$ gives

$$V(\sigma) := u''(\sigma) = \frac{1}{4} \psi_1(\sigma/2) + \sum_{p \in S} \frac{(\log p)^2 p^{-\sigma}}{(1 - p^{-\sigma})^2} > 0, \quad \partial_z \partial_{\bar{z}} \log h(\operatorname{Re} z) = \frac{V}{4}. \quad (5.4)$$

This physical norm is distinct from the squared holomorphic amplitude $|\Lambda_S(z)|^2$.

Proposition 5.1 (Determinant constraint). *For a finite-rank positive solution of (5.2), $\partial_z \partial_{\bar{z}} \log \det g = 0$. Consequently $\det g = |d(z)|^2$ locally for a nonvanishing holomorphic d . In particular, the determinant cannot equal $h(\operatorname{Re} z)^a |d(z)|^2$ for a nonzero real a .*

Proof. The finite-dimensional trace gives

$$\partial_z \partial_{\bar{z}} \log \det g = \text{tr } \mathcal{K} = \text{tr}[C, C^{\dagger_g}] = 0.$$

A real harmonic function is locally the real part of a holomorphic function, which proves the factorization. The proposed determinant instead has logarithmic curvature $aV/4 \neq 0$ by (5.4). $\square$

Thus tensoring a rank- N solution with the original arithmetic line creates determinant curvature $NV/4$. Prescribing h as one component is a different operation: $g = \text{diag}(h, h^{-1})$ is positive and has determinant one. Determinant cancellation is a necessary condition, not the full differential compatibility condition.

5.2 Ambient holomorphicity is a stronger restriction

Proposition 5.2 (Finite ambient-holomorphic obstruction). *Let $\mathcal{H}$ be a fixed positive Hilbert space and let $T(z) : \mathbb{C}^N \rightarrow \mathcal{H}$ be an injective Hilbert-norm holomorphic frame, with $N < \infty$. Set $g = T^*T$ and $P = Tg^{-1}T^*$, the orthogonal projector onto its span. If this induced metric satisfies (5.2), then P is constant on each connected parameter domain and $\mathcal{K} = 0$.*

Proof. With $T' = \partial_z T$, holomorphicity gives $\partial_z g = T^*T'$ and $\partial_{\bar{z}} g = T'^*T$. Hence

$$\mathcal{K} = -g^{-1}T'^*Tg^{-1}T^*T' + g^{-1}T'^*T' = g^{-1}T'^*(1 - P)T', \quad (5.5)$$

$$\text{tr } \mathcal{K} = \left\| (1 - P)T'g^{-1/2} \right\|_{\text{HS}}^2. \quad (5.6)$$

The trace must also vanish by Proposition 5.1. Thus $(1 - P)T' = 0$ and $\mathcal{K} = 0$. Direct differentiation yields $\partial_z P = (1 - P)T'g^{-1}T^* = 0$; taking the adjoint also gives $\partial_{\bar{z}} P = 0$. $\square$

This excludes a direct finite-dimensional holomorphic enlargement of the complete Morse–prime coherent family. In its fixed positive Hilbert space, that family is

$$\Psi_z = e^{zX/2} e^{-e^y/2}, \quad X = \frac{y}{2} - \sum_{p \in S} n_p \log p - \frac{\log \pi}{2}, \quad n_p \in \{0, 1, 2, \dots\}. \quad (5.7)$$

Projection to all $n_p = 0$ sends states at distinct positive real parameters σ_j to nonzero scalar multiples of $e^{\sigma_j y/4} e^{-e^y/2}$. These are linearly independent: division by $e^{-e^y/2}$ and successive differentiation at one point gives a Vandermonde system. Arbitrarily large independent sets exist on every open real interval. Consequently no constant finite-dimensional subspace can contain this family, as Proposition 5.2 would require.

The assumption $\partial_{\bar{z}} T = 0$ must not be substituted for the weaker physical condition $P\partial_{\bar{z}} T = 0$. An intrinsic holomorphic vacuum frame may have nonzero antiholomorphic derivatives in the massive complement. Such derivatives are allowed in physical tt^* geometry and invalidate the positive-trace calculation (5.6). Infinite rank likewise requires separate trace and domain arguments.

The corresponding line-bundle identity makes the remaining freedom explicit. For an intrinsic holomorphic section v , let $h_v = \langle v, v \rangle_g$ and $\beta = (1 - P_L)D_z v$, where P_L projects onto its line. In a holomorphic normal frame, differentiating $\log h_v$ gives

$$\partial_z \partial_{\bar{z}} \log h_v = \frac{\langle v, \mathcal{K}v \rangle_g + \|\beta\|_g^2}{h_v} = \frac{\|C^{\dagger_g} v\|_g^2 - \|Cv\|_g^2 + \|\beta\|_g^2}{h_v}. \quad (5.8)$$

Thus positive line curvature can be supplied by chiral mixing and variation inside a larger vacuum bundle. It is not itself a rank-two obstruction.

5.3 A local positive semisimple metric

Rank-two positivity and reality conditions are studied in [9]. The following explicit construction verifies the metric equations directly; it makes no existence assertion about a microscopic quantum theory.

Proposition 5.3 (Real-axis metric control). *There is a neighborhood of $z = 1/2$ carrying a positive rank-two solution of (5.2), with semisimple C_z , whose first diagonal component on the real axis is $\Lambda_S(\sigma)$.*

Proof. For $r = \operatorname{Re} w > 0$, set

$$g_*(w, \bar{w}) = \begin{pmatrix} \coth(2r) & 0 \\ 0 & \tanh(2r) \end{pmatrix}, \quad C_* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (5.9)$$

The first curvature entry equals the first commutator entry because

$$\frac{1}{4} \frac{d^2}{dr^2} \log \coth(2r) = \frac{4 \cosh(4r)}{\sinh^2(4r)} = \coth^2(2r) - \tanh^2(2r). \quad (5.10)$$

The second entries have opposite sign. The metric is positive with determinant one, while C_* has distinct eigenvalues ± 1 . For the constant bilinear pairing $\eta_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\eta_0 C_*$ is symmetric and $(\eta_0^{-1} g_*)(\eta_0^{-1} g_*) = 1$.

At $\sigma = 1/2$, $h > 1$ and $h' < 0$. Indeed $\Gamma(1/4) > \int_0^1 t^{-3/4}(1-t) dt = 16/5$, $\pi^{1/4} < 2$, and each Euler factor exceeds one. Also $\psi(1/4) = -\gamma - \pi/2 - 3 \log 2 < 0$, while the logarithmic derivatives of the Euler factors are negative. A local branch therefore exists for

$$w(z) = \frac{1}{4} \log \frac{\Lambda_S(z) + 1}{\Lambda_S(z) - 1}, \quad g_{\text{new}}(z, \bar{z}) = g_*(w(z), \overline{w(z)}), \quad C_z = w'(z) C_*. \quad (5.11)$$

After shrinking the neighborhood, $\operatorname{Re} w > 0$ and $w' \neq 0$ there. Holomorphic pullback multiplies both sides of (5.2) by $|w'|^2$. On the real axis $2w = \operatorname{arccoth} h$, so $(g_{\text{new}})_{11}(\sigma, \sigma) = \coth(2w(\sigma)) = h(\sigma)$. The eigenvalues of C_z are the distinct values $\pm w'(z)$. $\square$

The complex extension has changed: $(g_{\text{new}})_{11}(z, \bar{z})$ is generally not $h(\operatorname{Re} z)$. Only a real-axis diagonal norm has been retained. The construction does not specify the physical states (5.7), their complete overlap $\Lambda_S((\bar{z} + w)/2)$, a self-adjoint arithmetic observable X , or a Hamiltonian with the proposed metric as its vacuum metric. Semisimplicity and the displayed algebraic reality condition also do not establish the additional ultraviolet, infrared, and boundary data of a massive quantum field theory. The domain is local, with no all-prime or global half-plane continuation asserted.

Remark 5.4 (Keeping the original complex extension). For $g = \operatorname{diag}(h(\sigma), h(\sigma)^{-1})$ and $\kappa = V/4$, the companion choices $C = \begin{pmatrix} 0 & 1 \\ q & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 0 & q \\ 1 & 0 \end{pmatrix}$ require, respectively,

$$|q(z)|^2 = R_+(\sigma) := h^4 - \kappa h^2, \quad |q(z)|^2 = R_-(\sigma) := \kappa h^{-2} + h^{-4}.$$

Holomorphic q forces $RR'' - (R')^2 = 0$ wherever it is nonzero. Stirling's formula gives $u'' = 1/(2\sigma) + O(\sigma^{-2})$ and superexponential growth of h , whence $(\log R_+)'' = 2/\sigma + O(\sigma^{-2})$ and $(\log R_-)'' = -1/\sigma + O(\sigma^{-2})$ at infinity. Real analyticity on $(0, \infty)$ excludes either identity even on an open patch; identically zero q is excluded by the same large- σ behavior. This is a test of the stated frame and companion normalization, not every rank-two Higgs field. The changed extension in (5.11) avoids these hypotheses.

The control is deliberately permissive: any positive real-analytic norm greater than one with nonzero derivative can be inserted into the same local pullback. In particular, local metric compatibility does not select the arithmetic source normalization or its first normal logarithmic derivative. The next physical construction must preserve an explicitly defined complete pairing, rather than only one metric component, and derive its normalization from the theory. None of the metric results above supplies the Weil pole pairing or proves positivity of the complete Weil form.

6 Completion obstructions and vector-parameter geometry

The positive normalized arithmetic jump energy is an exact control, but its conversion into the Weil form requires a negative contact and a signed pole contribution. We first identify an obstruction to a scalar ground-state completion. We then construct a vector-parameter connection compatible with the benchmark Berry curvature and show why that compatibility does not fix the missing contact. Neither argument excludes positive bulk theories with more general source maps or internal degrees of freedom.

6.1 The complete form and its scalar ground transform

Fix $I = (-L/2, L/2)$ and a finite prime set $\mathcal{S}$ containing every prime with $\log p < L$. For zero extensions $F = E_L f$ and $G = E_L g$, set

$$\begin{aligned} n_\gamma(r) &= \frac{e^{-|r|/2}}{1 - e^{-2|r|}}, & a_{p,m} &= (\log p)p^{-m/2}, & d_{p,m} &= m \log p, \\ \nu_{\mathcal{S}}(dr) &= n_\gamma(r) dr + \sum_{p \in \mathcal{S}, m \geq 1} a_{p,m} (\delta_{d_{p,m}} + \delta_{-d_{p,m}})(dr), \end{aligned} \quad (6.1)$$

$$j(f, g) = \frac{1}{2} \int \nu_{\mathcal{S}}(dr) \int_{\mathbb{R}} \overline{F(x+r) - F(x)} [G(x+r) - G(x)] dx. \quad (6.2)$$

The gamma density is of order $1/(2|r|)$ at zero, and the prime measure has finite mass. Thus j is the nonnegative gamma form plus a bounded perturbation, with the same logarithmic Fourier form domain. Write J for its associated operator. The complete target has the exact form

$$W_L = J - \kappa_{\mathcal{S}} I + P_L, \quad \kappa_{\mathcal{S}} = -w_0 + 2 \sum_{p \in \mathcal{S}, m \geq 1} a_{p,m} > 0, \quad w_0 = \psi(1/4) - \log \pi, \quad (6.3)$$

where

$$p(x, y) = 2 \cosh((x - y)/2), \quad P_L = 2|c\rangle\langle c| - 2|s\rangle\langle s|, \quad c(x) = \cosh(x/2), \quad s(x) = \sinh(x/2). \quad (6.4)$$

Although $p(x, y)$ is positive, P_L is an indefinite rank-two operator. Inactive prime jumps contribute only a diagonal term to j , canceled by $\kappa_{\mathcal{S}}$; adding such primes does not change W_L .

Let $\mu_{\mathcal{S}}$ be the internal jump measure on I^2 obtained from (6.1), and let $k_{\text{esc}}(x) = \int_{x+r \notin I} \nu_{\mathcal{S}}(dr)$. Then j is the internal difference form plus $\int_I k_{\text{esc}} \bar{f}g$. Take a real $h \in C^1(\bar{I})$ bounded above and bounded away from zero. All identities below hold first for $f = hu$, $g = hv$ with $u, v \in C_c^\infty(I)$ and with zero exterior values for h in Jh . Expansion and symmetrization give

$$j(hu, hv) = \frac{1}{2} \int_{I^2} h(x)h(y) (\overline{u(x) - u(y)})(v(x) - v(y)) d\mu_{\mathcal{S}} + \int_I h(Jh) \bar{u}v. \quad (6.5)$$

Consequently the potential $-(Jh)/h$ admits a positive incidence-square representation. This is the useful scalar ground-state mechanism: a negative potential can arise from a positive energy after a

change of ground state. For example, $h = 1$ removes the positive exterior escape rate. Multiplying h by a constant, however, does not change $(Jh)/h$. The normalization of a vacuum ray cannot select the prescribed contact. The incidence operator can be closed to define a positive form; identifying that closure with a prescribed operator sum is a further domain question. Our obstructions use only the displayed compactly supported core.

The pole term has the corresponding identity

$$\begin{aligned} \langle f, P_L g \rangle &= \int_I r_P(x) \bar{f} g \\ &\quad - \frac{1}{2} \int_{I^2} p(x, y) (\overline{f(x) - f(y)}) (g(x) - g(y)) dx dy, \\ r_P(x) &= 8 \sinh(L/4) \cosh(x/2). \end{aligned} \tag{6.6}$$

Combining the two expansions yields the full polarized ground identity

$$\begin{aligned} q_L(hu, hv) &= \frac{1}{2} \int_{I^2} h(x) h(y) (\overline{u(x) - u(y)}) (v(x) - v(y)) [d\mu_S - p(x, y) dx dy] \\ &\quad + \int_I h(W_L h) \bar{u} v. \end{aligned} \tag{6.7}$$

Even a positive solution of $W_L h = 0$ would leave a signed edge measure. The poles change off-diagonal couplings and therefore cannot be treated merely as a local killing potential.

6.2 A sign obstruction before the first prime

Proposition 6.1 (Scalar source obstruction). *Let $\rho > 1$ be the root of $\rho^3 - \rho - 1 = 0$ and put $r_* = \log \rho$. For every $L > r_*$ there are nonzero nonnegative smooth inputs with disjoint supports for which $q_L(u, v) > 0$. Therefore, with a pointwise scalar source identification, q_L cannot be a positive-ground-state transform of a real scalar diffusion or jump form with nonnegative edge weights, even after adding an arbitrary local potential.*

Proof. Away from the finitely many active prime delays, the off-diagonal kernel is

$$R(r) = 2 \cosh(r/2) - n_\gamma(r) = e^{-r/2} \frac{e^{3r} - e^r - 1}{e^{2r} - 1}, \quad r > 0. \tag{6.8}$$

The polynomial $x^3 - x - 1$ is strictly increasing for $x > 1$, so R is positive precisely when $r > r_*$. Choose $d \in (r_*, L)$ away from all active prime delays and two sufficiently narrow patches separated by d . Their difference set avoids those delays and stays above r_* . For nonnegative u, v supported in the respective patches, $q_L(u, v) = \int u(x) R(|x - y|) v(y) dx dy > 0$. The contact and all local potentials vanish on these disjoint inputs. In contrast, a nonnegative scalar jump measure has a nonpositive cross term on disjoint nonnegative inputs; a local diffusion contributes zero. Positive multiplication by h preserves both disjointness and sign. $\square$

The restriction concerns a Markov-type scalar boundary energy, not positivity of an arbitrary Hilbert-space quadratic form. It also persists under scalar reversible bulk elimination. At a finite regulator a positive real symmetric diffusion/jump matrix has nonpositive off-diagonal entries. Its invertible interior block has entrywise nonnegative inverse, so the Schur complement again has nonpositive off-diagonal entries. Limits preserving the bilinear form on these test inputs retain the same sign.

Proposition 6.2 (Scalar phase obstruction). *For $L > 2r_*$, a scalar phase change cannot turn all the off-diagonal couplings into real nonpositive Markov couplings. This obstruction already occurs for some $L < \log 2$.*

Proof. Choose three narrow patches with all pairwise separations above r_* and away from prime delays. Their continuous couplings are positive. A phase change would have to assign phase -1 to each of the three edges. The product of phases around a triangle is 1 for a scalar gauge, whereas the required product is $(-1)^3 = -1$. The argument can be formulated almost everywhere on the patches and does not require pointwise regularity of the phase. Finally, evaluating $x^3 - x - 1$ at $\sqrt{2}$ shows $\rho < \sqrt{2}$, hence $2r_* < \log 2$. $\square$

Matrix states, genuine magnetic holonomy, and nonlocal or derivative source maps are outside these exclusions. The triangle identifies why scalar rephasing is insufficient; it does not exclude an energy containing interference squares with positive cross terms.

6.3 The compulsory diagonal and the contact problem

Indeed the pole kernel has a positive interference-square construction:

$$e_P^+(f, g) = \frac{1}{2} \int_{I^2} p(x, y) (\overline{f(x) + f(y)}) (g(x) + g(y)) dx dy = \langle f, P_L g \rangle + \int_I r_P \bar{f} g. \quad (6.9)$$

The equality follows simply by expanding the square. Its consequence is

$$q_L[f] = j[f] + e_P^+[f] - \int_I (\kappa_S + r_P(x)) |f(x)|^2 dx. \quad (6.10)$$

Thus a positive square reproduces the signed pole operator only together with an additional diagonal energy. The completion must derive the subtraction of this companion as well as the original contact. Adding two pole amplitudes does not by itself perform that task.

More precisely, put $D = \kappa_S I - P_L$. This bounded operator has infinite rank: it equals $\kappa_S I$ on $\{c, s\}^\perp$. A finite-rank correction of the prepared state map cannot remove the entire difference $J - W_L$. This does not exclude a finite number of bulk field species, each carrying infinitely many modes.

The danger of a circular Schur construction is already explicit in a valid one-prime support window. If $\log 2 < L \leq \log 3$, then $D > 0$: $c \perp s$, $\|c\|^2 = L/2 + \sinh(L/2)$, and $\|s\|^2 = -L/2 + \sinh(L/2)$, so its smallest possible exceptional eigenvalue is $\kappa_S - 2\|c\|^2 > 0$. Here $2\|c\|^2 \leq \log 3 + 2/\sqrt{3} < 3$, whereas $\kappa_S \geq -w_0 > 3$. The elementary square completion

$$\mathcal{A}[f, z] = j[f] - 2 \operatorname{Re} \langle D^{1/2} f, z \rangle + \|z\|^2 = q_L[f] + \|z - D^{1/2} f\|^2 \quad (6.11)$$

does have the desired minimized value. But $\mathcal{A} \geq 0$ holds if and only if $q_L \geq 0$. Its positivity is exactly the unknown assertion. Adding $\langle f, Df \rangle$ makes the action manifestly positive and changes its minimum to $j[f]$. A useful coherent constraint must therefore supply an independently derived infinite-dimensional contraction or projection identity, including the pole sectors, rather than repackaging the target's unknown positivity.

6.4 A vector-parameter lift and its normalization freedom

Vector-multiplet parameters offer a different compatibility equation from ordinary chiral tt^* geometry. Four-supersymmetry invariance constrains their Berry connection and projected moment-map operator

by a Bogomolny equation [5]. This is a necessary effective equation, not a converse construction of a microscopic theory.

For the finite-prime benchmark define, on $\text{Re } z > 0$,

$$\Lambda_S(z) = \pi^{-z/2} \Gamma(z/2) \prod_{p \in \mathcal{S}} (1 - p^{-z})^{-1}, \quad \ell(z) = \log \Lambda_S(z). \quad (6.12)$$

The logarithm exists and is real on the positive axis. The specified ground family has $\|\Psi_\sigma\|^2 = e^{\ell(\sigma)}$ and normalized extension $\Omega_{\sigma,\eta} = e^{i\eta X/2} \Omega_\sigma$. In the convention $A = i \langle \Omega, d\Omega \rangle$,

$$A_\sigma = 0, \quad A_\eta = -\frac{1}{2} \ell'(\sigma), \quad F_{\sigma\eta} = -\frac{1}{2} \ell''(\sigma), \quad \ell''(\sigma) = \text{Var}_\sigma(X) > 0. \quad (6.13)$$

Introduce an auxiliary third coordinate v , orient parameter space as (σ, η, v) , and define

$$A_\sigma = A_v = 0, \quad A_\eta = -\frac{1}{2} \text{Re } \ell'(\sigma + iv), \quad \Phi = -\frac{1}{2} \text{Im } \ell'(\sigma + iv). \quad (6.14)$$

The Cauchy–Riemann equations give $\nabla \times A = \nabla \Phi$: its σ and v components are, respectively, $-\text{Im } \ell''/2$ and $-\text{Re } \ell''/2$ on both sides. At $v = 0$ this restricts to (6.13), with $\Phi = 0$. This explicit half-space solution works for any suitable holomorphic ℓ real on the positive axis. It neither selects arithmetic factors nor makes the auxiliary extension the actual Berry data of a quantum model.

For completeness,

$$\ell'(z) = -\frac{\log \pi}{2} + \frac{1}{2} \psi(z/2) - \sum_{p \in \mathcal{S}} \frac{\log p}{p^z - 1}. \quad (6.15)$$

Its continued local-factor poles have logarithmic-derivative residue -1 , with multiplicities at coincidences. A simple boundary pole produces a two-dimensional dipole-type harmonic Φ , invariant along η , rather than a regular three-dimensional point monopole. Global defects and boundary conditions would require separate physical specifications.

The contact ambiguity remains exact. Replacing $\ell(z)$ by $\ell(z) + cz$, $c \in \mathbb{R}$, corresponds to rescaling the original source Ψ_ζ by $e^{c\zeta/2}$, where $\zeta = \sigma + i\eta$ labels that chosen complex family. Its normalized state changes only by $e^{ic\eta/2}$. Consequently A_η shifts by $-c/2$, a flat pure gauge on the noncompact η -line, while the curvature and Φ stay fixed. Nevertheless

$$\partial_\sigma \log |\Lambda_S(\sigma + i\tau)|^2 \mapsto \partial_\sigma \log |\Lambda_S(\sigma + i\tau)|^2 + 2c. \quad (6.16)$$

Thus curvature cannot determine this scalar contact. Changing the origin of X changes its mean but not its variance; a fixed physical source must supply the missing convention.

At $\sigma = 1/2$ the pole-free arithmetic multiplier is $2 \text{Re } \ell'(1/2 + i\tau) = -4A_\eta(1/2, \tau)$. This equality identifies functions, not a positive norm. The choice $v = \tau$ is an additional proposed correspondence, and the signed pole form is still absent. Scalar completion factors do not restore that form by a pointwise critical-line derivative. A successful enlargement must derive a physical source and the complete pairing, including its contact and residue or boundary contributions. These requirements leave matrix, fermionic, magnetic, and relative-boundary mechanisms open while specifying the precise failure of the scalar and curvature-only repairs tested here.

7 The narrowed physical-pairing problem

The results separate three requirements that a successful theory must satisfy together. It must have an ordinary positive Hilbert structure. It must retain the required arithmetic dependence in a specified observable. It must identify that observable, with its contact and poles, as the full norm of

a linear preparation map. The cylinder and prime theories satisfy the first requirement and parts of the second. None satisfies the third.

The interacting family in Section 4 is a useful exact control for the next calculation. Its mass parameter has no nonzero chiral class because it only dilates the fields. Changing the dimensionless interaction shape is a distinct deformation. For

$$G_c(A, B) = \frac{1}{2}(A^2 + B^2) + \frac{c}{4}(A^4 + B^4 + A^2B^2), \quad \mathcal{W}_c = G_c/2,$$

the superpotential variation is multiplication by

$$[\partial_c \mathcal{W}_c] = \left[\frac{1}{8}(A^4 + B^4 + A^2B^2) \right]$$

in the Jacobi ring. The Higgs operator for the differential (4.8) is half this variation. Its critical values are not all equal. Deriving its matrix geometry, the physical source frame and the boundary conditions is therefore a concrete next task, rather than a change of variables already covered by the dilation argument.

The reference period and physical norm define the comparison

$$\mathcal{R}(c, \bar{c}) = \frac{H_c}{|C_c|^2} \tag{7.1}$$

where the period is nonzero. A common source rescaling cancels from this ratio. It can distinguish a physical normalization relation from a removable frame factor, but its physical interpretation requires care: the real-cycle holomorphic integral is not already a normalizable boundary state. A gluing law would need to construct the relevant boundary state and establish how its positive Gram pairing relates to (7.1). Neither a topological intersection form nor a signed index can be substituted for that Hermitian pairing.

If a shape $c = c(M)$ is introduced along the prime mass, then the period and metric become $C(c(M))/M$ and $H(c(M), \bar{c}(\bar{M}))/|M|^2$. The compensating normalization needed to retain an Euler factor must be fixed by the physical boundary problem, rather than chosen after inspecting the desired arithmetic answer. A protected relation controlling the period and norm together would be substantive progress even before a complete Weil realization is available.

The periodic gamma sector presents a second concrete task. Its physical vacuum exists, but its arithmetic source-frame norm has not been calculated. The multivalued insertion $-Y/2$ requires deck, flavor or equivalent global parameter data. Identifying the relevant physical gamma pairing could determine whether this sector supplies more than the existing holomorphic period.

Finally, the source space cannot be omitted. A nine-dimensional vacuum fiber cannot represent all smooth compactly supported inputs. Boundary fields, periodic sectors, excitations or another specified infinite-dimensional construction must supply the full preparation map and the prime returns. The scalar sign test directs attention to internal interference or nonlocal observables, while the compulsory diagonal identity gives an immediate check of any proposed pole mechanism. The next theory should derive its complete bilinear response before large numerical overlap calculations are undertaken.

7.1 Reproducibility and limitations

The accompanying small programs verify nine common-core supercharge identities and two rational-algebra identities in the quartic period elimination. They use exact integer or rational arithmetic and Python's standard library. Their records do not verify self-adjoint domains, the analytic cycle

hypotheses, the Hodge theorem or the full Weil form; those arguments are stated separately in the text. No floating-point positivity sweep or large numerical dataset enters the results.

The source package includes a manuscript build recipe, a deliverable manifest and a replay validator. Large future regenerable outputs are assigned to the external `szp-archive` layout, with source, parameters and small hash records retained in the repository. The conclusions of this draft are the explicit constructions and scoped obstructions above. The full positive arithmetic pairing remains the research objective.

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